

Predictive Construction Project Overrun Model																									
Executive early-warning view for cost overruns and schedule delays. All projects and results are synthetic.																									
0.899		0.774		62.5%		81.1%																			
Cost ROC-AUC		Cost PR-AUC		Cost Precision		Cost Recall																			
0.756		0.524		52.0%		45.3%																			
Schedule ROC-AUC		Schedule PR-AUC		Schedule Precision		Schedule Recall																			
<table> <tr> <th>Risk Band</th><th>Test Projects</th></tr> <tr> <td>Red</td><td>188</td></tr> <tr> <td>Yellow</td><td>138</td></tr> <tr> <td>Green</td><td>110</td></tr> </table>		Risk Band	Test Projects							Red	188	Yellow	138	Green	110	Test Risk Distribution <table border="1"> <caption>Test Risk Distribution Data</caption> <thead> <tr> <th>Risk Band</th> <th>Count</th> </tr> </thead> <tbody> <tr> <td>Red</td> <td>188</td> </tr> <tr> <td>Yellow</td> <td>138</td> </tr> <tr> <td>Green</td> <td>110</td> </tr> </tbody> </table>								Risk Band	Count
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#	Finding	Evidence																							
1	Cost prediction	The cost-overrun champion achieved ROC-AUC 0.899 and PR-AUC 0.774 on the 2025 test set.																							
2	Schedule prediction	Schedule ROC-AUC decreased to 0.756 on the 2025 test set, creating a model-monitoring priority.																							
3	Portfolio screening	The test set contains 188 Red, 138 Yellow, and 110 Green projects.																							
4	Human review	Predictions identify review priority; they do not authorize autonomous project decisions.																							
5	Feature lineage	The model combines Project 1 cost/schedule controls with Project 2 RFI and change-workflow indicators.																							
Project ID	Project	Band	Overall	Cost	Schedule	Drivers																			
PRJ3-01418	Gateway Bridge Program 1418	Red	99.8%	94.0%	97.4%	Slow change approval; Workflow risk score; Overdue RFI backlog																			
PRJ3-01705	Granite Urban Village 1705	Red	99.8%	89.1%	97.7%	Overdue RFI backlog; Slow change approval; Workflow risk score																			
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PRJ3-01716	Redwood Logistics Center 1716	Red	99.2%	91.6%	90.9%	Overdue RFI backlog; Slow RFI response; Workflow risk score																			
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PRJ3-01492	Horizon Education Center 1492	Red	98.6%	92.1%	82.4%	Overdue RFI backlog; Low CPI; Low SPI																			
PRJ3-01752	Lakeview Highway Improvements 1752	Red	98.5%	87.1%	88.3%	Overdue RFI backlog; Low CPI; Low SPI																			
PRJ3-02338	Riverside Neighborhood 2338	Red	98.2%	74.5%	92.9%	Overdue RFI backlog; Slow change approval; Slow RFI response																			

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Synthetic-data disclosure: All project records and model outcomes are fictional. Model performance is not an industry benchmark, and production use on real construction projects is not authorized by this portfolio project.

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1. Executive Summary

This case study is the third project in a construction analytics portfolio. It combines the cost and schedule controls developed in Project 1 with the RFI and change-workflow indicators developed in Project 2 to predict future cost overruns and schedule delays from early and mid-project snapshots.

Dataset	Cost model	Schedule model	Test risk bands
2,362 clean projects 1,577 train 349 validation 436 test	Random Forest ROC-AUC 0.899 PR-AUC 0.774	Logistic Regression ROC-AUC 0.756 PR-AUC 0.524	188 Red 138 Yellow 110 Green

Principal conclusions

- Cost-overrun classification generalized strongly to the future synthetic test period.
- Schedule-delay performance decreased from validation to test, creating a real model-monitoring concern rather than a result to hide.
- CPI and current cost variance were the strongest cost-overrun predictors.
- SPI, planned completion, procurement delay, and overdue workflow indicators were important schedule predictors.
- Predictions were converted into Red, Yellow, and Green review priorities, but every score requires human review.
- The Act phase defines a controlled-pilot path; it does not authorize production use.

2. Business Problem and Portfolio Progression

Primary business question

Which early and mid-project indicators best predict material construction cost overruns and schedule delays, and how can management prioritize intervention before the outcomes become unavoidable?

Portfolio progression

Project	Title	Contribution
Project 1	Construction Project Controls Analytics	CPI, SPI, cost, schedule, contingency, EAC, and early-warning controls
Project 2	Construction Change Order and RFI Analytics	RFI response, change approval, pending exposure, revision loops, and forecast lag
Project 3	Predictive Construction Project Overrun Model	Time-validated probability and magnitude prediction with governance

Project 3 uses a new independent synthetic dataset. It references feature concepts from Projects 1 and 2 without mixing their folders, repositories, or records.

3. Analytical Lifecycle and Governance

Phase	CRISP-DM	DMAIC	Microsoft	PMI governance
Ask	Business Understanding	Define	Business Understanding	Initiating / Planning
Prepare	Data Understanding	Measure	Data Acquisition and Understanding	Planning
Process	Data Preparation	Measure / early Analyze	Data Acquisition and Understanding	Executing / Monitoring

Phase	CRISP-DM	DMAIC	Microsoft	PMI governance
Analyze	Modeling and Evaluation	Analyze	Modeling	Executing / Monitoring
Share	Evaluation	Analyze / improvement design	Customer Acceptance	Stakeholder engagement
Act	Deployment	Improve and Control	Deployment / Acceptance	Executing / Monitoring / Closing

The frameworks are conceptually aligned but are not treated as identical. The workflow is iterative, and quality or governance findings may require returning to earlier phases.

4. Dataset, Feature Lineage, and Quality

The deterministic generator created 2,400 intended projects with snapshots between approximately 35% and 60% planned completion. Raw tables intentionally contained duplicates, category variants, missing critical values, invalid numeric values, invalid dates, target inconsistencies, and orphan relationships.

Control	Result
Raw project rows	2,412
Raw workflow rows	2,417
Raw outcome rows	2,416
Duplicate removals	36
Quarantined records	123
Clean model rows	2,362
Cleaning actions	199
SQLite integrity	ok
Foreign-key violations	0

Feature families

- Project and contract attributes
- CPI, SPI, cost variance, schedule variance, contingency, and productivity
- Procurement, weather, and safety indicators
- RFI response, overdue rates, revisions, reopening, and RFI-to-change conversion
- Approved and pending change exposure, approval cycle, old pending rate, and forecast incorporation lag
- Workflow handoffs, revision loops, and snapshot workflow-risk score

Final outcomes, final CPI/SPI, actual end date, final EAC, and final status were excluded from predictor matrices.

5. Modeling Methodology

Element	Method
Time split	Train 2019–2023; Validation 2024; Test 2025
Cost classification	Final cost overrun $\geq 10\%$
Schedule classification	Final schedule delay ≥ 30 days
Cost regression	Final cost-overrun percentage
Schedule regression	Final schedule-delay days
Classification candidates	Logistic Regression and Random Forest
Regression candidates	Ridge Regression and Random Forest Regressor

Element	Method
Champion selection	Validation PR-AUC for classification; validation MAE for regression
Threshold selection	Maximum validation F1; test untouched
Explainability	Permutation importance on held-out test data

The model uses 40 predictors: 31 numeric and 9 categorical. Numeric fields are standardized for linear models, and categorical values are one-hot encoded with unknown-category handling.

6. Classification Performance

Target	Champion	ROC-AUC	PR-AUC	Precision	Recall	F1	Brier
Cost overrun	Random Forest	0.899	0.774	0.625	0.811	0.706	0.123
Schedule delay	Logistic Regression	0.756	0.524	0.520	0.453	0.484	0.183

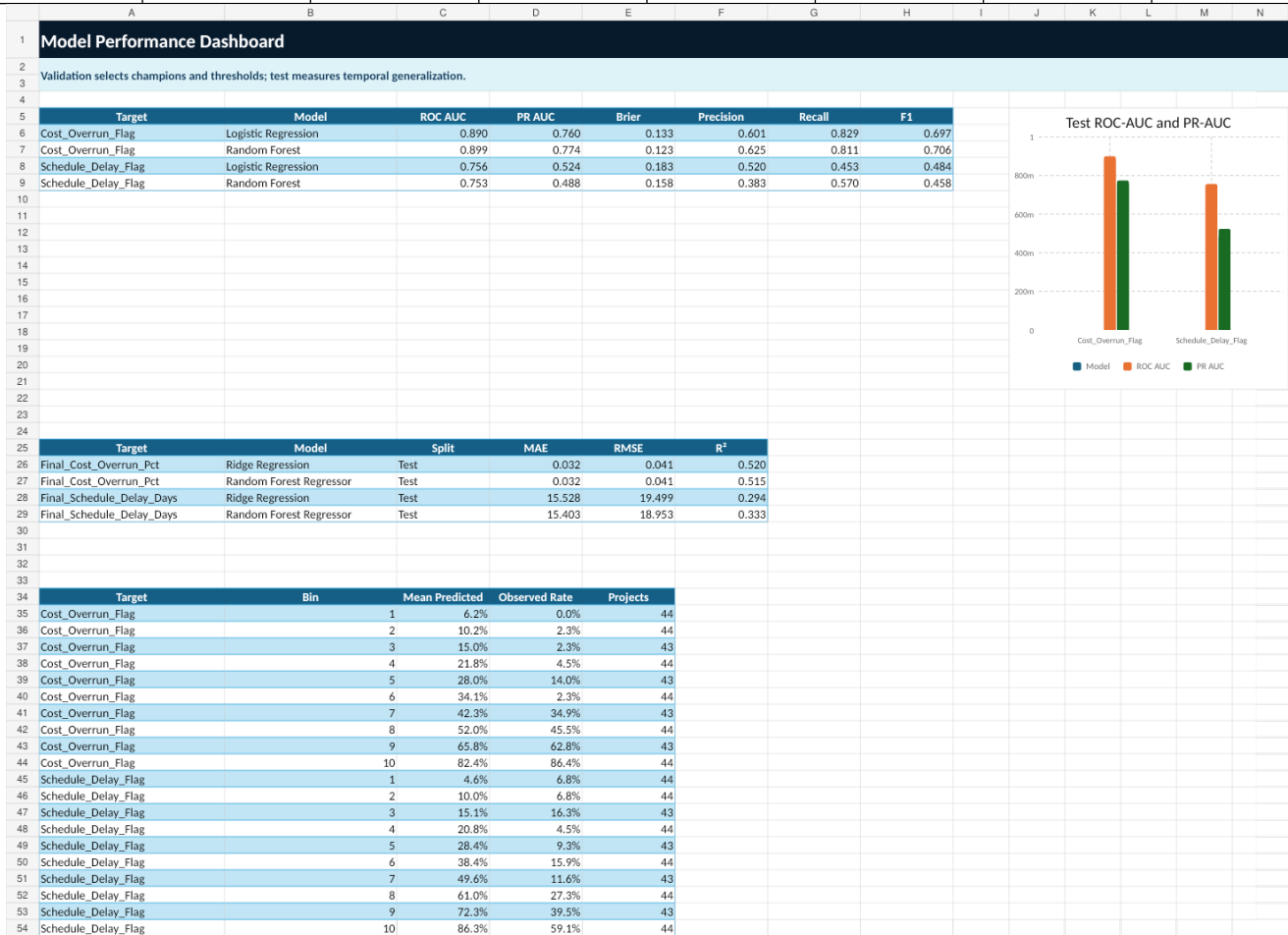


Figure 1. Classification, regression, and calibration performance.

7. Regression, Calibration, and Temporal Generalization

Target	Model	MAE	RMSE	R ²	Interpretation
Final_Cost_Overrun_Pct	Ridge Regression	0.032	0.041	0.520	Cost-overrun percentage points

Target	Model	MAE	RMSE	R ²	Interpretation
Final_Cost_Overrun_Pct	Random Forest Regressor	0.032	0.041	0.515	Cost-overrun percentage points
Final_Schedule_Delay_Days	Ridge Regression	15.528	19.499	0.294	Schedule-delay days
Final_Schedule_Delay_Days	Random Forest Regressor	15.403	18.953	0.333	Schedule-delay days

Temporal degradation

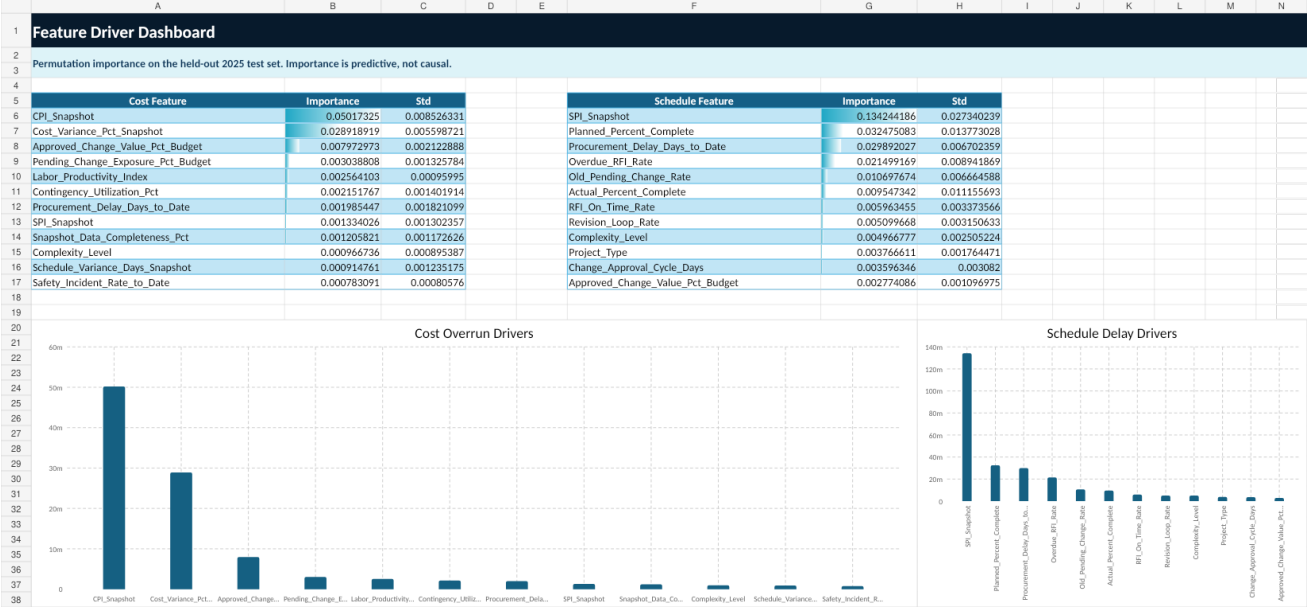
The schedule classifier achieved validation ROC-AUC 0.856 but test ROC-AUC 0.756. The reduction suggests temporal variation in the synthetic portfolio. The result is retained as an example of why model performance must be monitored after deployment.

The monitoring demonstration produced a schedule probability PSI of 0.105, above the proposed Yellow threshold of 0.10.

8. Feature Drivers and Project Risk Scoring

Rank	Cost feature	Importance	Schedule feature
1	CPI_Snapshot	0.0502	SPI_Snapshot
2	Cost_Variance_Pct_Snapshot	0.0289	Planned_Percent_Complete
3	Approved_Change_Value_Pct_Budget	0.0080	Procurement_Delay_Days_to_Date
4	Pending_Change_Exposure_Pct_Budget	0.0030	Overdue_RFI_Rate
5	Labor_Productivity_Index	0.0026	Old_Pending_Change_Rate
6	Contingency_Utilization_Pct	0.0022	Actual_Percent_Complete
7	Procurement_Delay_Days_to_Date	0.0020	RFI_On_Time_Rate
8	SPI_Snapshot	0.0013	Revision_Loop_Rate
9	Snapshot_Data_Completeness_Pct	0.0012	Complexity_Level
10	Complexity_Level	0.0010	Project_Type

Feature importance measures predictive contribution under permutation. It does not establish that changing a feature will cause a better outcome.



Top predicted-risk projects

Project	Band	Overall	Cost	Schedule	Primary drivers
Gateway Bridge Program 1418	Red	99.8%	94.0%	97.4%	Slow change approval; Workflow risk score; Overdue RFI backlog
Granite Urban Village 1705	Red	99.8%	89.1%	97.7%	Overdue RFI backlog; Slow change approval; Workflow risk score
Horizon Neighborhood 34	Red	99.4%	80.3%	96.8%	Overdue RFI backlog; Workflow risk score; Slow RFI response
Redwood Logistics Center 1716	Red	99.2%	91.6%	90.9%	Overdue RFI backlog; Slow RFI response; Workflow risk score
Aspen Corporate Center 702	Red	99.1%	91.8%	89.2%	Overdue RFI backlog; Slow change approval; Slow RFI response
Northstar Highway Improvements 1925	Red	99.0%	88.1%	92.0%	Slow change approval; Workflow risk score; Slow RFI response
Pioneer Urban Village 1981	Red	98.8%	94.1%	80.4%	Overdue RFI backlog; Slow RFI response; Workflow risk score

9. Dashboards and Portfolio Communication

The Share phase produced executive, model-performance, project-risk, and feature-driver dashboards plus Power BI and Tableau build specifications.

	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3											
4											
	Rank	Project ID	Project	Type	Complexity	Owner Decision	Cost Prob.	Schedule Prob.	Overall Prob.	Band	Drivers
6	1	PRJ3-01418	Gateway Bridge Program 1418	Heavy Civil	Very High	Slow	94.0%	97.4%	99.8% Red		Slow change approval; Workflow risk score; Overdue RFI backlog
7	2	PRJ3-01705	Granite Urban Village 1705	Mixed-Use	Very High	Slow	89.1%	97.7%	99.8% Red		Overdue RFI backlog; Slow change approval; Workflow risk score
8	3	PRJ3-00034	Horizon Neighborhood 34	Residential	Very High	Slow	80.3%	96.8%	99.4% Red		Overdue RFI backlog; Workflow risk score; Slow RFI response
9	4	PRJ3-01716	Redwood Logistics Center 1716	Industrial	Very High	Slow	91.6%	90.9%	99.2% Red		Overdue RFI backlog; Slow RFI response; Workflow risk score
10	5	PRJ3-00702	Aspen Corporate Center 702	Commercial	High	Slow	91.8%	89.2%	99.1% Red		Overdue RFI backlog; Slow change approval; Slow RFI response
11	6	PRJ3-01925	Northstar Highway Improvements 1925	Heavy Civil	Very High	Slow	88.1%	92.0%	99.0% Red		Slow change approval; Workflow risk score; Slow RFI response
12	7	PRJ3-01981	Pioneer Urban Village 1981	Mixed-Use	Very High	Slow	94.1%	80.4%	98.8% Red		Overdue RFI backlog; Slow RFI response; Workflow risk score
13	8	PRJ3-01492	Horizon Education Center 1492	Institutional	High	Slow	92.1%	82.4%	98.6% Red		Overdue RFI backlog; Low CPI; Low SPI
14	9	PRJ3-01752	Lakeview Highway Improvements 1752	Heavy Civil	Moderate	Slow	87.1%	88.3%	98.5% Red		Overdue RFI backlog; Low CPI; Low SPI
15	10	PRJ3-02338	Riverside Neighborhood 2338	Residential	High	Slow	74.5%	92.9%	98.2% Red		Overdue RFI backlog; Slow change approval; Slow RFI response
16	11	PRJ3-00088	Harbor District 88	Mixed-Use	Very High	Fast	90.7%	80.2%	98.2% Red		Overdue RFI backlog; Slow RFI response; Workflow risk score
17	12	PRJ3-01155	Copper Education Center 1155	Institutional	High	Slow	70.0%	93.4%	98.0% Red		Slow RFI response; Slow change approval; Workflow risk score
18	13	PRJ3-01906	Liberty Education Center 1906	Institutional	Very High	Fast	51.0%	95.0%	97.5% Red		Overdue RFI backlog; Slow RFI response; Slow change approval
19	14	PRJ3-00192	Cedar Research Facility 192	Institutional	High	Slow	84.6%	82.8%	97.4% Red		Overdue RFI backlog; Workflow risk score; Slow change approval
20	15	PRJ3-02161	Gateway Utility Corridor 2161	Infrastructure	Very High	Slow	84.1%	82.1%	97.2% Red		Overdue RFI backlog; Slow RFI response; Low SPI
21	16	PRJ3-00084	Gateway Earthworks Package 84	Heavy Civil	High	Standard	58.1%	93.0%	97.0% Red		Overdue RFI backlog; Low SPI; Low CPI
22	17	PRJ3-01281	Evergreen District 1281	Mixed-Use	High	Standard	87.0%	74.9%	96.7% Red		Overdue RFI backlog; Workflow risk score; Low CPI
23	18	PRJ3-01248	Evergreen Residences 1248	Residential	High	Slow	81.8%	81.2%	96.6% Red		Overdue RFI backlog; Low SPI; Low CPI
24	19	PRJ3-00260	Riverside Bridge Program 260	Heavy Civil	High	Slow	80.8%	81.4%	96.4% Red		Overdue RFI backlog; Slow change approval; Workflow risk score
25	20	PRJ3-00404	Cedar Logistics Center 404	Industrial	High	Slow	88.3%	69.4%	96.4% Red		Overdue RFI backlog; Slow change approval; Workflow risk score
26	21	PRJ3-02204	Mesa Civic Complex 2204	Institutional	Very High	Slow	85.1%	75.7%	96.4% Red		Slow change approval; Workflow risk score; Overdue RFI backlog
27	22	PRJ3-00641	Canyon Retail Center 641	Commercial	High	Slow	90.8%	59.9%	96.3% Red		Overdue RFI backlog; Low CPI; Pending change exposure
28	23	PRJ3-01519	Granite Utility Corridor 1519	Infrastructure	Very High	Standard	47.1%	92.9%	96.2% Red		Slow change approval; Overdue RFI backlog; Slow RFI response
29	24	PRJ3-01875	Canyon Housing 1875	Residential	Moderate	Slow	48.3%	87.6%	96.1% Red		Slow change approval; Overdue RFI backlog; Low CPI
30	25	PRJ3-01379	Summit Apartments 1379	Residential	Very High	Standard	71.0%	86.0%	95.9% Red		Overdue RFI backlog; Low SPI; Low CPI
31	26	PRJ3-02104	Summit Transit Program 2104	Infrastructure	Moderate	Standard	72.6%	85.2%	95.9% Red		Overdue RFI backlog; Low CPI; Low SPI
32	27	PRJ3-02151	Crescent Town Center 2151	Mixed-Use	High	Standard	69.0%	86.2%	95.7% Red		Slow change approval; Low CPI; Low SPI
33	28	PRJ3-01769	Summit Town Center 1769	Mixed-Use	Very High	Standard	73.6%	83.6%	95.7% Red		Low CPI; Low SPI; Pending change exposure
34	29	PRJ3-01363	Redwood Utility Corridor 1363	Infrastructure	Moderate	Slow	71.4%	84.7%	95.6% Red		Slow change approval; Slow RFI response; Workflow risk score
35	30	PRJ3-00929	Liberty District 929	Mixed-Use	High	Slow	86.1%	68.5%	95.6% Red		Overdue RFI backlog; Low SPI; Low CPI

The communication design shows probability and risk band together, labels validation and test periods, and keeps synthetic-data, uncertainty, and human-review warnings visible.

10. Act Plan, Human Review, and Monitoring

Controlled-pilot roadmap

- Days 0–30: approve charter, prohibited uses, independent review, reproducibility, source mapping, and data-quality gates.
- Days 31–60: conduct a silent pilot, train reviewers, and review thresholds and monitoring.
- Days 61–90: conduct a limited visible pilot, record decisions and overrides, and make a formal go/no-go decision.

Human review

1. Verify input data and snapshot context.
2. Review cost, schedule, and overall probabilities.
3. Review model and rule-based drivers.
4. Consider current project context and known events.
5. Accept, Override, Escalate, or Defer.
6. Assign an action, owner, due date, and evidence.
7. Record later outcomes for monitoring.

Proposed monitoring

Metric	Yellow	Red / stop
Cost ROC-AUC	Yellow <0.82	Red <0.78
Schedule ROC-AUC	Yellow <0.75	Red <0.70
Probability PSI	Yellow >0.10	Red >0.20
Critical data quality	Any failure	Stop scoring
Segment AUC gap	Yellow >0.10	Red >0.15
Override rate	Yellow >30%	Red >50%
Model age	Review at 12 months	Mandatory decision at 18 months

11. Limitations, Responsible AI, and Conclusion

- All data is synthetic and may not represent real project distributions.
- Performance is not an industry benchmark.
- Target thresholds are case-study definitions.
- Feature importance is predictive, not causal.
- The schedule model shows temporal degradation.
- Segment results have limited sample sizes.
- Real deployment requires approved data, independent validation, calibration, drift review, human oversight, access controls, and incident response.
- The model may not be used for contractual, legal, safety-critical, or personnel decisions.

Conclusion

Project 3 completes a three-project construction analytics portfolio progression from descriptive project controls, to diagnostic workflow analytics, to predictive early warning. The project demonstrates technical modeling and business governance together, including the discipline to retain unfavorable temporal results and restrict unvalidated production use.

12. Sources and Deliverables

Professional references

IBM CRISP-DM overview: <https://www.ibm.com/docs/en/spss-modeler/18.6.0?topic=dm-crisp-help-overview>

ASQ DMAIC: <https://asq.org/quality-resources/dmaic>

Microsoft Data Science Lifecycle: <https://learn.microsoft.com/en-us/shows/dev-intro-to-data-science/what-is-the-data-science-lifecycle-2-of-28>

PMI Process Groups: <https://www.pmi.org/standards/process-groups>

Primary project deliverables

- Analysis workbook and model-performance tables
- Saved classification and regression pipelines
- Feature manifest and model card
- Executive and operational dashboards
- Power BI and Tableau specifications
- Batch scoring and model-monitoring scripts
- Controlled-pilot Act plan, risk register, RACI, and override log
- Complete Ask–Prepare–Process–Analyze–Share–Act documentation

End of Report